Hypothesis Testing: Two Samples

MDragon Data Tools - Study Guide

Comparing Two Population Means

When to Use

- Comparing two groups (treatment vs control, before vs after, etc.)
- Independent samples OR paired/dependent samples

Independent Samples T-Test

Conditions

1. Random, independent samples
2. Approximately normal distributions OR n_1, $n_2 \geq 30$
3. Equal or unequal variances

Test Statistic (Equal Variances)

```
t = (x̄₁ - x̄₂) / √[s²ₚ(1/n₁ + 1/n₂)]

Pooled variance: s²ₚ = [(n₁-1)s₁² + (n₂-1)s₂²] / (n₁ + n₂ - 2)
df = n₁ + n₂ - 2
```

Example: Independent Samples

📝 **Problem:** Group A (n=20): $\bar{x}_1=85$, $s_1=10$. Group B (n=25): $\bar{x}_2=80$, $s_2=12$. Test if means differ ($\alpha=0.05$).

💡 **Solution:**

1. $H_0: \mu_1 = \mu_2$; $H_a: \mu_1 \neq \mu_2$
2. $s^2_p = [19(100) + 24(144)] / 43 = 5356/43 = 124.56$
3. $t = (85-80) / \sqrt{[124.56(1/20 + 1/25)]} = 5 / 3.35 = 1.49$
4. df = 43, p-value ≈ 0.143
5. 0.143 > 0.05, **Fail to reject H_0**

Conclusion: No significant difference between groups.

Paired Samples T-Test

When to Use

- Before/after measurements
- Matched pairs
- Same subjects measured twice

Test Statistic

```
t = d̄ / (sᵈ / √n)

where:
d̄ = mean of differences
sᵈ = standard deviation of differences
df = n - 1
```

Example: Paired Samples

📝 **Problem:** Weight loss program. 8 people: before={180,175,200,165,190,185,195,170}, after={175,170,195,162,185,180,190,165}. Test if program works (α=0.05).

💡 **Solution:**

1. Differences: d = {5,5,5,3,5,5,5,5}
2. $\bar{d} = 4.75$, $s_d = 0.886$
3. $H_0: \mu_d = 0$; $H_a: \mu_d > 0$ (right-tailed)

4. $t = 4.75 / (0.886/\sqrt{8}) = 15.16$, df=7
5. p-value ≈ 0.000001
6. **Reject H_0**

Conclusion: Program significantly reduces weight.

Two Proportion Z-Test

Test Statistic

```
z = (p̂₁ - p̂₂) / √[p̄(1-p̄)(1/n₁ + 1/n₂)]

Pooled proportion: p̄ = (x₁ + x₂) / (n₁ + n₂)
```

Example

📝 **Problem:** Survey: 60/100 men prefer product A, 45/80 women prefer it. Test if proportions differ ($\alpha=0.05$).

💡 **Solution:**

1. $\hat{p}_1 = 0.60$, $\hat{p}_2 = 0.5625$
2. $\bar{p} = 105/180 = 0.583$
3. $z = 0.0375 / 0.074 = 0.507$
4. p-value = 0.612
5. **Fail to reject H_0**

Use our Hypothesis Test Calculator for quick analysis!
